MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING

MODULE 2 UNIT 1
Designing an agentic AI tech stack

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Learning outcome:

LO1: Examine how agentic AI is operationalized through scaffolding and orchestration.

1. Introduction

In this module, you will explore how agentic AI is implemented in practice. You will examine how agents are structured, workflows are orchestrated, and leadership decisions shape deployment. The focus is on turning capability into execution, moving from models that generate text to systems that act reliably inside organizations.

2. Orchestrating agents at scale

In this video, Jacob Andreas introduces a critical extension to agent architecture: the ability to learn over time. Without it, agents repeat the same reasoning steps for every interaction, limiting efficiency and scalability.

He outlines three forms of learning:

1. **User memory:** Building profiles that allow agents to personalize future interactions
2. **Procedural memory:** Capturing successful workflows so agents can reuse them without recomputing each step
3. **Model-level learning:** Using patterns from many interactions to improve the system itself over time

The underlying insight is that memory is not about storing everything, but about selectively capturing what matters in a usable, shareable form.

Main message: Agents become operationally valuable when they can learn from experience and reduce repeated work over time.

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Video 1: Orchestrating agents at scale. (Access this set of notes on the Online Campus to engage with this video and download its transcript.)

In Video 2, Andreas shifts the conversation from capability to risk. As agents become more powerful, failure modes expand in two directions: agents making mistakes and users exploiting them.

- On the system side, **language models remain unpredictable**. They can hallucinate, execute flawed logic, or take actions without clear user consent. These are not edge cases but structural realities of probabilistic systems.
- On the user side, **malicious prompting** introduces a second layer of risk. Users can manipulate agents into violating rules or triggering unintended actions.

The key insight is architectural; you cannot rely on the model to manage these risks. Safeguards must sit outside the model. Critical actions require explicit human approval. Sensitive data must never be exposed by default. Monitoring systems must flag high-risk behavior.

Main message: Robust agentic systems are not just intelligent. They are controlled, constrained, and designed to fail safely.

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Video 2: Human and agent teams. (Access this set of notes on the Online Campus to engage with this video and download its transcript.)

3. Agent abilities

Andreas concludes this series of videos by looking at where agentic systems are heading next. He highlights multi-agent systems, with a controller agent that can delegate work to specialized subagents. In some cases, this is best understood as one larger agentic system with distributed parts. In others, especially across organizations, it raises more complex questions about coordination, access, and what information agents should share on behalf of users.

He also turns to multimodal and physical-world agents. As models become able to read images, generate outputs, and write code that controls tools, agents can begin to interact beyond text, including in robotics. But Andreas is careful not to overstate the frontier. These systems still depend on lower-level controllers and remain limited in fine-grained perception, manipulation, and safety.

Main message: The frontier is expanding quickly, but agent design still comes back to the five fundamentals of observation, action, planning, memory, and safety.

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Video 3: Agent abilities. (Access this set of notes on the Online Campus to engage with this video and download its transcript.)

Note:

At this point in the lesson, you can test your understanding of the content with a practice quiz. Access this lesson on the Online Campus to begin.

4. Conclusion

The main theme of this module is that the difference between a language model and an agent is not scale, but scaffolding. Across architecture, leadership, and construction, you see the same system from different angles. Agents observe, act, plan, remember, and operate within safety constraints. Tools enable action. Memory allows learning. Orchestration turns isolated steps into workflows.

At the leadership level, implementation is a set of choices. Whether an organization elects to build, buy, or combine agent offerings depends on autonomy and capability. Human oversight is calibrated through thresholds, rather than eliminated. Speed matters because short cycles drive learning and flexibility matters because locking into a stack creates risk.

From a construction perspective, complex tasks are decomposed, coordinated, and executed through agentic loops. Methodologies become workflows. Workflows become systems.

Agentic AI works when systems are deliberately designed. The model is only one component. The surrounding architecture determines whether it delivers value.

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